Computational Macro – Problem Set 2

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1

1.1

a) (a, z) is the state because it is all that we need to know to determine our total resources available today. Using $c = z + Ra - a'$ to get rid of c to have a' as our only choice variable, the Bellman equation reads

$$v(a, z) = \max_{a'} \{u(z + Ra - a') + \beta \mathbb{E}\left[v(a', z') \mid z\right]\}$$

b)

$$\begin{aligned} \frac{\partial v(a,z)}{\partial a'}: \quad & -u_c(z + Ra - a') + \beta \frac{\partial \mathbb{E}\left[v(a', z') \mid z\right]}{\partial a'} = 0 \\ \iff \quad & -u_c(z + Ra - a') + \beta \mathbb{E}\left[v_a(a', z') \mid z\right] = 0 \\ \iff \quad & u_c(c) = \beta \mathbb{E}\left[v_a(a', z') \mid z\right] \end{aligned}$$

c) To derive the envelope condition for $v_a(a, z)$, we begin by plugging the optimal policy function $a' = g(a, z)$ into the Bellman equation. As $g(a, z)$ is the optimal choice that achieves the optimum, we drop the $\max_{a'}$ operator:

$$v(a, z) = u(z + Ra - g(a, z)) + \beta \mathbb{E}\left[v(g(a, z), z') \mid z\right]$$

Differentiating both sides with respect to a, we obtain

$$v_a(a, z) = (R - g_a(a, z))\, u_c(z + Ra - g(a, z)) + g_a(a, z)\, \beta \mathbb{E}\left[v_a(g(a, z), z') \mid z\right]$$

Using $a' = g(a, z)$, $c = z + Ra - g(a, z)$, and the optimality condition found in (b), we

have

$$\begin{aligned} & v_a(a,z) = (R - g_a(a,z))\, u_c(c) + g_a(a,z)\, u_c(c) \\ \iff \quad & v_a(a,z) = R u_c(c) \end{aligned}$$

That is, the envelope condition; the marginal value of an additional unit of assets today is the gross return R times the marginal utility of consumption today.
d) Now, since $v_a(a,z) = Ru_c(c)$ holds for any state, we have $v_a(a',z') = Ru_c(c')$. Subbing this in the expression found in (b), we have

$$u_c(c) = \beta R\, \mathbb{E}\left[u_c(c') \mid z\right]$$

That is, the desired form for the Euler equation.

1.2

a) Given $u(c) = -\frac{1}{\gamma}e^{-\gamma c}$, we have $u_c(c) = e^{-\gamma c}$. Plugging this into the general Euler equation derived in 1.1 (d), we obtain the CARA Euler equation $e^{-\gamma c} = \beta R \mathbb{E}\left[e^{-\gamma c'} \mid x, e\right]$. Now, substituting our conjecture for c into $a' = x - c$, we have $a' = x - \mu x - \nu e - f = (1-\mu)x - \nu e - f$. Moreover, we can write tomorrow's consumption conjecture as $c' = \mu x' + \nu e' + f$, where tomorrow's cash on hand is $x' = z' + Ra' = \bar{z} + e' + Ra'$ and tomorrow's income deviation is $e' = \rho e + \sigma_\varepsilon \varepsilon'$, so that we can write

$$\begin{aligned} c' &= \mu\bar{z} + (\mu+\nu)e' + \mu R a' + f \\ &= \mu\bar{z} + (\mu+\nu)(\rho e + \sigma_\varepsilon \varepsilon') + \mu R a' + f \end{aligned}$$

We substitute in the expression we found for a', obtaining

$$c' = \mu\bar{z} + (\mu+\nu)(\rho e + \sigma_\varepsilon \varepsilon') + \mu R\left((1-\mu)x - \nu e - f\right) + f$$

Rearranging terms, we obtain

$$c' = \mu R(1-\mu)x + \left[\mu(\rho - R\nu) + \nu\rho\right] e + \mu(\bar{z} - Rf) + f + (\mu+\nu)\sigma_\varepsilon \varepsilon'$$

that is, the desired expression.

b) In the Euler equation $e^{-\gamma c} = \beta R \mathbb{E}\left[e^{-\gamma c'} \mid x, e\right]$, the term inside the expectation operator takes the form e^Y, where $Y = -\gamma c'$. We know from the problem text that, conditionally on (x, e), c' is normally distributed with mean equal to the sum of the first four terms of (2) and variance equal to $(\mu+\nu)^2\sigma_\varepsilon^2$. Thus, for $Y = -\gamma c'$, we have $\mathbb{E}[Y \mid x, e] = -\gamma \mathbb{E}[c' \mid x, e]$ and $\text{Var}[Y \mid x, e] = (-\gamma)^2 \text{Var}[c' \mid x, e] = \gamma^2(\mu+\nu)^2\sigma_\varepsilon^2$.

We now apply the formula for $\mathbb{E}\left[e^Y\right]$. Substituting it in our Euler equation, we obtain

$$e^{-\gamma c} = \beta R \exp\left(-\gamma \mathbb{E}[c' \mid x, e] + \frac{\gamma^2(\mu+\nu)^2\sigma_\varepsilon^2}{2}\right)$$

Taking logs, this expression becomes

$$-\gamma c = \ln(\beta R) - \gamma \mathbb{E}[c' \mid x, e] + \frac{\gamma^2(\mu+\nu)^2\sigma_\varepsilon^2}{2} \iff c = \mathbb{E}[c' \mid x, e] - \frac{\ln(\beta R)}{\gamma} - \frac{\gamma(\mu+\nu)^2\sigma_\varepsilon^2}{2}$$

Now, we sub in the LHS our conjecture $c = \mu x + \nu e + f$ and in the RHS the expected value $\mathbb{E}[c' \mid x, e]$, that is, the first four terms of (2):

$$\mu x + \nu e + f = \mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]\, e + \mu(\bar{z} - Rf) + f - \frac{\ln(\beta R)}{\gamma} - \frac{\gamma(\mu+\nu)^2\sigma_\varepsilon^2}{2}$$

Now, to find the coefficients of (3), we equalize coefficients on both sides:

• equalizing coefficients on x, we have $\mu = \mu R(1-\mu) \iff \mu[1 - R(1-\mu)] = 0$, an equation of degree 2 with two solutions. We can exclude the solution $\mu = 0$: if it were $\mu = 0$, then in the constant term $\mu(\bar{z} - Rf) + f$ the first term with f would disappear, and matching coefficients on e would give $\nu = \nu\rho$, i.e. $\nu = 0$. The constant equation would then reduce to $\ln(\beta R) = 0$, which fails unless $\beta R = 1$; and even then f would be undetermined, and consumption would not respond to wealth, so assets would grow without bound, violating the transversality condition. So, as $\mu \neq 0$ we can divide the expression we found by μ, obtaining thus $1 = R(1-\mu)$. Using $R = 1 + r$, this can be written as $\mu = \frac{r}{R}$.

We move to equalizing the coefficients on e:

$$\nu = \mu(\rho - R\nu) + \nu\rho \iff \nu(1 + \mu R - \rho) = \mu\rho \iff \nu = \frac{\mu\rho}{1 + \mu R - \rho}$$

Using $\mu = \frac{r}{R}$, where $r = R - 1$, we have $1 + \mu R = 1 + \left(\frac{R-1}{R}\right) R = 1 + R - 1 = R$ and so

the expression for ν can be written as $\nu = \frac{\mu\rho}{R-\rho}$.

Finally, equalizing the constant terms on the two sides of the Euler equation we have $f = \mu(\bar{z} - Rf) + f - \frac{\ln(\beta R)}{\gamma} - \frac{\gamma(\mu+\nu)^2\sigma_\varepsilon^2}{2}$. Subtracting f from both sides and using $\mu = \frac{r}{R}$, this becomes

$$rf = \mu\bar{z} - \frac{\ln(\beta R)}{\gamma} - \frac{\gamma(\mu+\nu)^2\sigma_\varepsilon^2}{2} \iff f = \frac{\mu}{r}\bar{z} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+\nu)^2\sigma_\varepsilon^2}{2r}$$

$$\iff \quad f = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+\nu)^2\sigma_\varepsilon^2}{2r}$$

We have thus obtained all the desired expressions for point (b).

c) From the consumption rule $c = \mu x + \nu e + f$, we have $\mu = \frac{\partial c}{\partial x}$; μ is the derivative of consumption with respect to the available resources, that is, the marginal propensity to consume.

Now, we have $\mu = \frac{r}{R} = \frac{r}{1+r}$. This means that if resources increase by one unit today, the household will consume a share $\mu = \frac{r}{1+r}$ today and save the remaining share $1 - \mu = \frac{1}{1+r}$. When we multiply the saved fraction by the gross interest rate $R = 1 + r$, we obtain $\frac{1}{1+r} \cdot (1 + r) = 1$; this means that tomorrow's cash on hand x' will increase by 1 full unit, returning the household's wealth to the same +1 position it had today. Tomorrow, it can again consume $\frac{r}{R}$ and will keep the extra 1 unit of wealth for the following period, repeating this for all periods.

In line with the Permanent Income Hypothesis, the household smooths consumption over time by consuming only the permanent interest flow generated by an extra unit of wealth, μ, leaving the principal untouched so that consumption remains permanently higher by $\frac{r}{R}$ in every future period.

d) As we have $e = \rho e_{-1} + \sigma_\varepsilon\varepsilon$ and $x = \bar{z} + e + Ra$, if an innovation raises e by 1 unit, it raises e directly by 1, and it also raises x by 1. Hence, from $c = \mu x + \nu e + f$, the total effect on c will be the sum of the partial derivatives with respect to x and e, that is, $\mu + \nu$. Substituting in $\mu = \frac{r}{R}$ and $\nu = \frac{\rho\mu}{R-\rho}$, we have

$$\mu + \nu = \frac{r}{R} + \frac{\rho\mu}{R-\rho} = \frac{r}{R} + \frac{\rho\frac{r}{R}}{R-\rho} = \frac{r}{R}\left(1 + \frac{\rho}{R-\rho}\right) = \frac{r}{R} \cdot \frac{R}{R-\rho} = \frac{r}{R-\rho}$$

We can see that as the persistence ρ increases, the denominator $R - \rho$ decreases, and so the expression increases.

At the limit $\rho \to 0$, the expression equals $\frac{r}{R} = \mu$; as in this case the income shock is purely

transitory and carries no information about future income, to smooth consumption the household will save most of this windfall increase in income and only consume its annuity value $\mu = \frac{r}{R}$.

At the limit $\rho \to 1$, the expression goes to $\frac{r}{r} = 1$; as this shock is so persistent that the household is guaranteed to have higher income in every future period, it will decide to consume all this extra income today.

e) When $\rho = 0$, our expressions from point (b) become $\nu = 0$ and

$$f = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma \mu^2 \sigma_\varepsilon^2}{2r} = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma r \sigma_\varepsilon^2}{2R^2}$$

whereas $\mu = \frac{r}{R}$ does not depend on ρ and thus stays unchanged. Plugging these parameters back into $c = \mu x + \nu e + f$ and $a' = x - c$, we obtain

$$c(x) = \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma r \sigma_\varepsilon^2}{2R^2} \quad \text{and} \quad a'(x) = \frac{1}{R}x - \frac{\bar{z}}{R} + \frac{\ln(\beta R)}{\gamma r} + \frac{\gamma r \sigma_\varepsilon^2}{2R^2}$$

Note that with $\rho = 0$, i.e., iid income, e drops out as a separate state variable – when $\rho = 0$, z provides no information about z', so the household only needs to track x.

2

2.1

Comparing the analytical coefficients from Part 1 to the exact coefficients of the problem discretized via Rouwenhorst on $N_z = 7$ grid points, we find:

- Analytical (Gaussian): $\mu = 0.038462$, $\nu = 0.247253$, $f = 0.95766091$
- Discretized (Rouwenhorst): $\mu = 0.038462$, $\nu = 0.247189$, $f_{N_z} = 0.95766213$

The marginal propensity to consume out of cash on hand, $\mu = r/R$, is identical across both specifications because it does not depend on the distribution of the income shock. The coefficient on the income state ν differs by about 6.4×10^{-5}, and the precautionary intercept f differs by about 1.22×10^{-6}. These differences are negligible and not economically meaningful, as the Rouwenhorst method matches the conditional mean and variance

of the AR(1) Gaussian process closely, and higher-order moments have a minimal impact for $\sigma = 0.1$.

3

3.1

(a) Under CARA, $u(c) = -\frac{1}{\gamma}e^{-\gamma c}$, marginal utility is $u_c(c) = e^{-\gamma c}$, and its inverse is $u_c^{-1}(m) = -\frac{1}{\gamma}\ln(m)$; both u and u_c are defined for all real consumption levels $c \in \mathbb{R}$ (allowing negative consumption in principle), while $u_c^{-1}(m)$ is defined for $m > 0$. Under CRRA (with $\gamma = 2$), $u(c) = \frac{c^{1-\gamma}}{1-\gamma} = -c^{-1}$, marginal utility is $u_c(c) = c^{-\gamma}$, and its inverse is $u_c^{-1}(m) = m^{-1/\gamma}$; because of the Inada condition $\lim_{c\to 0^+} u_c(c) = \infty$, u and u_c are strictly defined only for positive consumption $c > 0$ (and u_c^{-1} for $m > 0$).

(b) In the CARA model, the Rouwenhorst chain discretizes the persistent deviation e in levels (with unconditional standard deviation $\sigma = 0.1$), and income in levels is obtained additively as $z_m = \bar{z} + e_m$. In the CRRA model, the chain discretizes log-income $y = \ln z$ (using unconditional standard deviation $\sigma_y = \sigma_\epsilon/\sqrt{1-\rho^2}$), and strictly positive income in levels is obtained by exponentiating the grid nodes: $z_m = \exp(y_m)$.

(c) In the CARA model without an explicit borrowing constraint, the lower bound is set wide enough ($a_1 = -15$) so as not to bind on the window of interest, and the grid is equidistant (linear). In the CRRA model, the lower bound is pinned down exactly by the exogenous borrowing limit $a_1 = a_{\min} = 0$. The upper bound a_{N_a} is pinned down by impatience ($\beta R < 1$): at high wealth levels, the impatience motive dominates the precautionary saving motive, ensuring that net savings become negative ($a' < a$) and assets stay bounded inside the grid. The grid points should be dense near the borrowing constraint $a_1 = 0$ (using quadratic spacing), where the consumption policy function exhibits the strongest curvature and a kink.

(d) In the CARA model, only the artificial grid bounds $a_1 \le a' \le a_{N_a}$ restrict tomorrow's assets, so admissible consumption at state (a_i, z_m) with cash on hand $x_{im} = z_m + Ra_i$ is $c_{im} \in [x_{im} - a_{N_a},\, x_{im} - a_1]$. In the CRRA model, the borrowing

limit $a' \geq 0$ (which coincides with $a_1 = 0$), the upper grid bound $a' \leq a_{N_a}$, and the strict positivity of consumption ($c > 0$) jointly restrict choices: maximum consumption is $c_{im}^{\max} = x_{im} - a_1 = x_{im}$ (when the borrowing constraint $a' = 0$ binds), and minimum consumption is $c_{im}^{\min} = \max\{x_{im} - a_{N_a}, \epsilon_c\}$ for a small positive threshold $\epsilon_c > 0$ so that marginal utility remains finite.